

KIRAN M. JAYASURYA

Scientist - ISRO · PhD Research Fellow - RRI

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PROFESSIONAL SUMMARY

Scientist in the Space Astronomy Group at the U R Rao Satellite Centre (URSC), ISRO, and PhD researcher at the Raman Research Institute (RRI). Specializing in X-ray binaries, pulsars, and ultraluminous X-ray sources. Proven track record in broadband X-ray spectroscopy, photoelectric polarimetry, and the development of space-based X-ray astronomy payloads for future ISRO missions.

RESEARCH INTERESTS

Compact Objects: X-ray binaries, pulsars, and ultraluminous X-ray sources.

High-Energy Phenomena: Tidal disruption events around supermassive black holes, gamma-ray bursts, and multi-messenger astronomy.

Techniques & Instrumentation: Broadband X-ray spectroscopy, photoelectric polarimetry, and space-based payload development.

PROFESSIONAL APPOINTMENTS

Scientist, Space Astronomy Group

Jan 2021 – Present

U R Rao Satellite Centre (URSC), ISRO, Bengaluru

- Leading the design, simulation, and development of space-based astronomy payloads for upcoming ISRO missions.
- Conducting advanced spectro-polarimetric analysis of galactic and extragalactic X-ray sources using multi-mission observatory data.

EDUCATION

PhD in Astronomy and Astrophysics

2025 – Present

Raman Research Institute (RRI), Bengaluru

M.Sc. in Astronomy and Astrophysics

2018 – 2020

Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram

B.Tech. in Engineering Physics

2015 – 2018

Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram

RESEARCH EXPERIENCE

Postgraduate Researcher

2018 – 2020

Indian Institute of Space Science and Technology (IIST)

- Modeled and analyzed tidal disruption events around supermassive black holes to understand late-time emission mechanisms.

Student Researcher

2017 – 2020

Indian Institute of Space Science and Technology (IIST)

- Conducted comprehensive gamma-ray burst (GRB) afterglow modeling to extract prompt emission and kinetic parameters.

Student Researcher

Jun 2018

Physical Research Laboratory (PRL), Ahmedabad

- Investigated the physical properties and dynamics of accretion disks around compact objects.

TECHNICAL PROFICIENCIES

- **Programming & Scripting:** Python, MATLAB, C++, L^AT_EX
- **Simulation & Detector Modeling:** Geant4, Garfield++, COMSOL Multiphysics
- **Astronomy Data Analysis:** HEASoft, CIAO, XMM-Newton SAS, HENDRICS

SELECTED PUBLICATIONS

2026

- Jayasurya, K. M., Upadhyay, A., & Rana, V. (2026). "Discovery of a Candidate 2 keV Cyclotron Resonance Scattering Feature in the HLX NGC 3583 X-1." *The Astrophysical Journal*, 1007(1), 32.
- Majumder, S., Kushwaha, A., Singh, S., Jayasurya, K. M., Das, S., & Nandi, A. (2026). "Probing the accretion geometry of black hole X-ray binaries: a multimission spectro-polarimetric and timing study." *Monthly Notices of the Royal Astronomical Society*, 545(2), staf1933.
- Ghosh, A., Razzaque, S., Barnard, J., et al. (including K. M. Jayasurya). (2026). "Photometric and late-time spectropolarimetric observations of GRB 250129A afterglow." *Monthly Notices of the Royal Astronomical Society*, 551(2), stag1391.
- Ghosh, A., Dimple, Misra, K., et al. (including K. M. Jayasurya). (2026). "Exploring the multi-wavelength properties of the high energetic event ZTF20abbiixp/GRB 200524A." arXiv:2608.09704.

2025

- Chatterjee, R., Vadodariya, K., Vatedka, R., Agrawal, V. K., Tyagi, A., Jayasurya, K. M., et al. (2025). "XSPECT on-board XPoSat: calibration and first results." *Journal of Astronomical Telescopes, Instruments, and Systems*, 11(4), 044007.
- Vatedka, R., Tyagi, A., Vadodariya, K., Agrawal, V. K., Chatterjee, R., Chellammal, R. M., Jayasurya, K. M., et al. (2025). "X-ray spectroscopy and timing experiment on XPoSat: instrument configuration and science prospects." *Journal of Astronomical Telescopes, Instruments, and Systems*, 11(3), 035001.

2024 & 2023

- Majumder, S., Chatterjee, R., Jayasurya, K. M., Das, S., & Nandi, A. (2024). "First Detection of X-Ray Polarization in Galactic Ultraluminous X-Ray Pulsar Swift J0243.6+6124 with IXPE." *The Astrophysical Journal Letters*, 971(1), L21.
- Jayasurya, K. M., Agrawal, V. K., & Chatterjee, R. (2023). "Detection of significant X-ray polarization from transient NS-LMXB XTE J1701–462 with IXPE and its implication on the coronal geometry." *Monthly Notices of the Royal Astronomical Society*, 525(3), 4657–4662.
- Kushwaha, A., Jayasurya, K. M., Agrawal, V. K., & Nandi, A. (2023). "IXPE and NICER view of black hole X-ray binary 4U 1630–47..." *MNRAS: Letters*, 524(1), L15–L20.
- Chatterjee, R., Agrawal, V. K., Jayasurya, K. M., & Katoch, T. (2023). "Spectro-polarimetric view of bright atoll source GX 9+9 using IXPE and AstroSat." *MNRAS: Letters*, 521(1), L74–L78.

PROCEEDINGS & CONFERENCES

Other Publications

- Jayasurya, K. M., Vadodariya, K., & Radhakrishna, V. (2024). "Multifaceted Simulations to design a TPC Polarimeter for Soft to Medium X-rays." *42nd meeting of the Astronomical Society of India*, 42, P11.
- Surabhi, C., Resmi, L., & Jayasurya, K. M. (2024). "Exploring Gamma-Ray Burst Radio Afterglow Population." *42nd meeting of the Astronomical Society of India*, 42, P137.
- Majumder, S., Chatterjee, R., Jayasurya, K. M., Das, S., & Nandi, A. (2024). "First IXPE view of ULX pulsar Swift J0243.6+6124: Polarimetric results and detection of pulsation." *45th COSPAR Scientific Assembly*, 45, 1586.
- Perley, D., Copperwheat, C., Guolo-Pereira, M., et al. (2020). "Follow-up observations of GSNST-20ca (AT2020rml) with the Liverpool Telescope." *Transient Name Server AstroNote*, 161, 1.

Workshops & Symposiums

- 2025 — XSPECT-XPoSat Workshop, URSC, Bengaluru
- 2025 — Astronomical Society of India, NIT Rourkela
- 2024 — NICER-IXPE Workshop, Washington, D.C.
- 2024 — National Space Science Symposium, University of Goa, Panaji
- 2024 — Astronomical Society of India, IISc Bengaluru
- 2020 — GROWTH Time Domain Astronomy Workshop, California Institute of Technology